DERIVED LOWER SHRIEK ALONG A THICKENING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Topological invariance of the small etale topos (Etale Cohomology, Theorem etale-cohomology-theorem-topological-invariance, canonical tag 04DZ). cited at proof lines 3922-3925. Gives that i^{-1} and i_* = i_! are inverse equivalences for the thickening i.
- Composition of lower shriek functors (more-etale, Lemma lemma-shriek-composition, canonical tag 0F7J). cited at proof line 3927. Identifies Rf_! with Rf'_! composed with Ri_! for the composite f = f' after i.

Source reference. Original theorem 04DZ: sources/stacks-chow/tags.txt, source lines 5745–5745. Cited at proof lines 3922-3925: makes i^{-1} and i_* = i_! inverse equivalences for the thickening i.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at proof line 3927: identifies Rf_! = Rf'_! composed with Ri_! for f = f' after i.

Lemma 0.2. *Let $f' : X' \to Y$ be a finite type separated morphism of quasi-compact and quasi-separated schemes. Let $i : X \to X'$ be a thickening and denote $f = f' \circ i$. Let Λ be a ring. For K' in $D^+_{tors}(X'_{\acute{e}tale}, \Lambda)$ or $K' \in D(X'_{\acute{e}tale}, \Lambda)$ if Λ is torsion, we have $Rf_! i^{-1} K' = Rf'_! K'$.*

Proof. This is true because i^{-1} and $i_* = i_!$ inverse equivalences of categories by the topological invariance of the small étale topos (Étale Cohomology, Theorem 04DZ) and we can apply Lemma 0F7J. □

Date: stacks-project:more-etale.tex:local-snapshot-20261012; GNU FDL 1.2 or later.